

Table S4. Exploratory cross-window recommendation-adherence associations

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Within participant, linked cycle				
Sleep	-0.26 (-0.95 to 0.43)	0.453	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Pre-sleep	-1.09 (-2.75 to 0.56)	0.258	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Pre-sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Between participants, observed monitoring-period average				
Sleep	-2.52 (-3.55 to -1.48)	<0.001	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 2.52 percentage points lower average Sleep adherence over the monitoring period.
Pre-sleep	-3.59 (-5.94 to -1.25)	0.005	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 3.59 percentage points lower average Pre-sleep adherence over the monitoring period.

All four tests form one FDR-correction set. Site-standardized estimates give nine sites equal weight and use 50:50 Work/Free weighting. The primary sample contained 1,376 target-window rows from 140 participants and 761 linked cycles. Within-participant claims are withheld because temporal dependence remains unresolved. Between-participant estimates describe observed monitoring-period averages, not rankings, stable traits, or causal effects. Confidence intervals are not multiplicity adjusted.

Supplementary Table S4. Exploratory within-participant and between-participant associations across Brown et al. recommendation windows. Results are percentage-point differences per 10 percentage points higher Daytime adherence. The four tests form one FDR family. The within-participant day-level claim is withheld because serial dependence remains unresolved. Between-participant results describe observed monitoring-period averages and do not establish rankings, stable traits, or causal effects.